

Rami L.

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EDUCATION

The University of Texas at Austin

Austin, TX

Bachelor of Science in Electrical and Computer Engineering

Graduating Spring 2028

- **Relevant Coursework:** Digital System Design with HDLs, Digital Logic Design, Software Design and Implementation I, Intro to Embedded Systems, Circuit Theory, Linear Systems and Signals, Information Security and Privacy
- **Leadership & Activities:** Texas Longhorn Band Piccolo, Kappa Kappa Psi Service Fraternity, Filipino Student Association, IEEE, Student Engineers Educating Kids

WORK EXPERIENCE

R&D Engineering Intern – Embedded Systems & Product Testing

Summer 2026

Clairvoyant Networks, Inc.

- Developed embedded firmware and supported hardware redesign for a restroom/bedroom monitoring system, integrating sensor interfacing, serial configuration, Wi-Fi microcontroller research, and low-power design considerations for a next-generation prototype.
- Implemented and debugged UART-based serial configuration features, enabling real-time threshold tuning, sensor output monitoring, and validation of fall, flush, and leak detection behavior.
- Calibrated embedded fall-detection performance across restroom and bedroom environments by tuning firmware threshold parameters, validating sensor placement, and verifying fall-from-bed detection with dual bed sensor setups.

Radio Systems Engineer

Spring 2025 – Spring 2026

Longhorn Lemons / Orange Dames Racing

- Reviewed preliminary circuit design concepts for a student-built endurance race car, focusing on wiring layout, power distribution, and system reliability.
- Contributed to early electrical planning discussions with mechanical and design subteams.

Teaching Assistant, ECE 111S - The University of Texas at Austin

Fall 2026

The University of Texas at Austin

Austin, TX

- Support first-year ECE students by answering academic questions, facilitating seminar activities, and connecting students with engineering resources.

PROJECTS

Synchronous FIFO Buffer | *SystemVerilog, Verilog, Xilinx Vivado*

Summer 2026

- Designed a parameterized synchronous FIFO with read/write pointer control, full/empty flag generation, and configurable data width/depth.
- Developed self-checking SystemVerilog testbenches to verify data ordering, overflow/underflow handling, and simultaneous read/write behavior.

Programmable Stopwatch/Timer (Digital Logic Design Project) | *Verilog, Basys3 FPGA*

Spring 2026

- Designed a programmable stopwatch/timer on a Basys3 FPGA using Verilog RTL, supporting count-up/count-down operation and externally loaded preset values.
- Built a controller/datapath architecture with HLSM-based design planning, mode selection logic, start/stop control, reset behavior, and 10 ms timing resolution across four 7-segment displays.

Space Invaders (Embedded Systems Project) | *C, MSPM0, ADC/DAC, ST7735R LCD*

Spring 2025

- Interfaced, wired, and programmed a game system with switches, a slide-pot joystick using ADC input, speaker audio using DAC output, and an LCD connected to a Texas Instruments MSPM0 microcontroller.
- Programmed start and end screens with English and Spanish language options and implemented game sprites in C using ST7735R TFT drivers.

TECHNICAL SKILLS

Languages: C/C++, Python, Java, HTML/CSS, Verilog/SystemVerilog LC-3/ARM Assembly

Tools: GitHub, Visual Studio Code, Xilinx Vivado, KiCad, MATLAB, Tera Term, Excel/Google Sheets